CS 59300
Application of Deep Learning
Machine Learning

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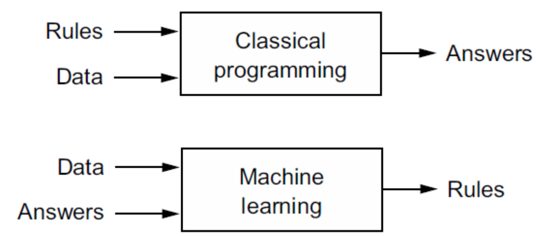
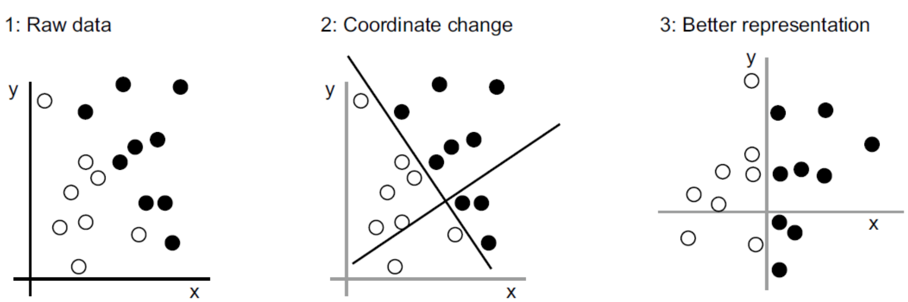
What is Machine Learning (ML)?

- Could a general-purpose computer "originate" anything?
- Find unknown target function: $f: x \rightarrow y$
- An ML system is trained rather than explicitly programmed.



Machine Learning (ML)

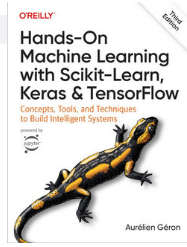
- ML models are all about finding appropriate representations for their input data — transformations of the data that make it more amenable to the task at hand.



Machine Learning (ML)

- What is learning in ML?
- Learning describes an automatic search process for data transformations that produce useful representations of some data, guided by some feedback signal — representations that are amenable to simpler rules solving the task at hand.

Outline



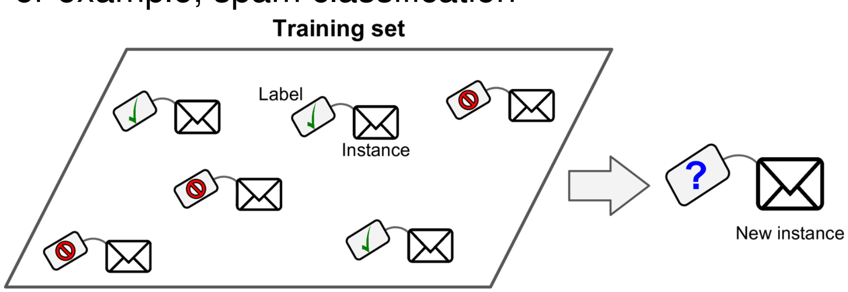
- Type of machine learning
- Key components for machine learning
- End-to-end machine learning project

Type of machine learning

- Five major categories
  - Supervised learning
  - Unsupervised learning
  - Semisupervised learning
  - Self-supervised learning
  - Reinforcement learning

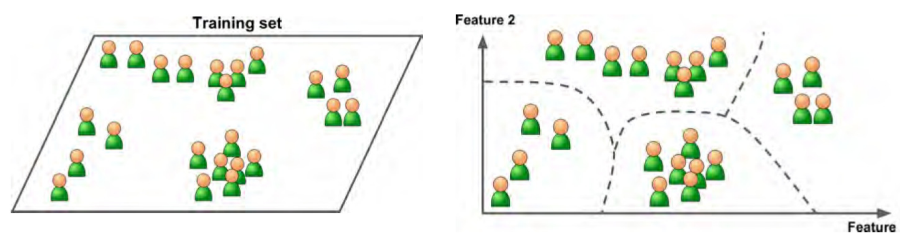
Supervised Learning

- The training set fed to the machine learning includes the desired solutions, called labels
- Learn with a "teacher"
- Our main focus
- For example, spam classification



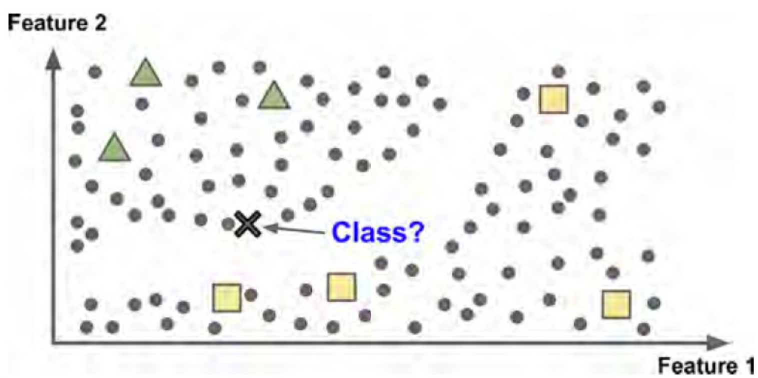
Unsupervised Learning

- The training data is unlabeled.
- The system tries to learn without a teacher.



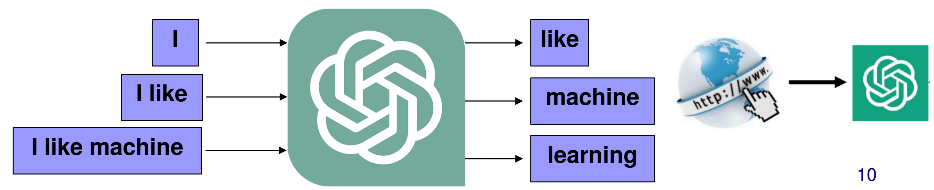
Semisupervised Learning

- The training data are partially labeled.



Self-supervised Learning

- (From Wikipedia page) A model is trained on a task using the data itself to generate supervisory signals, rather than relying on external labels provided by humans.
- Example: ChatGPT training
  - Text solitaire



Reinforcement Learning

- The learning system, called an agent in this context, can observe the environment, select and perform actions, and get rewards in return (or penalties in the form. of negative rewards).
- It must then learn by itself what is the best strategy, called a *policy*, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.

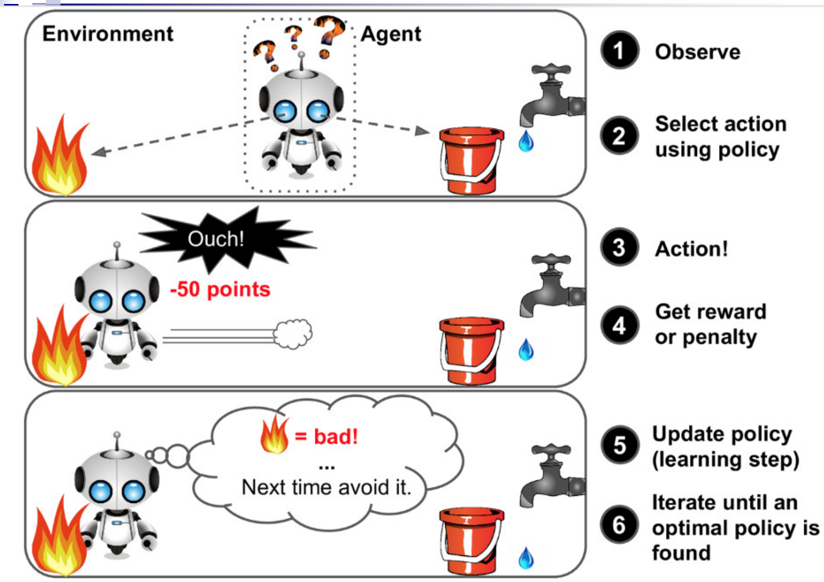


Figure 1-12. Reinforcement Learning

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Categories Based on Output

- Two types
  - Classification
    - Predict a class (e.g., digits 0, 1, …, 9)
  - Regression
    - Predict a value (e.g., house prices)

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Outline

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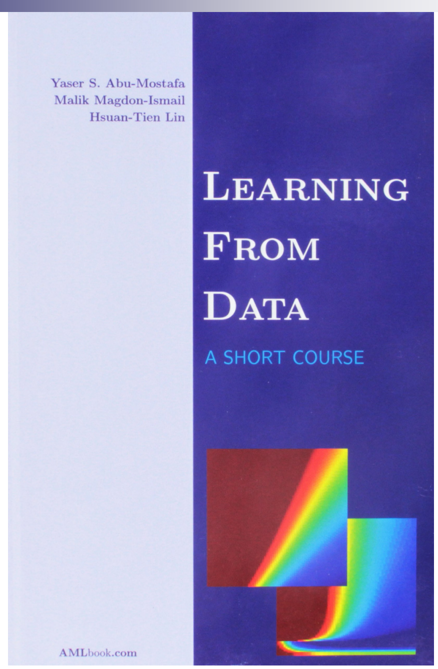

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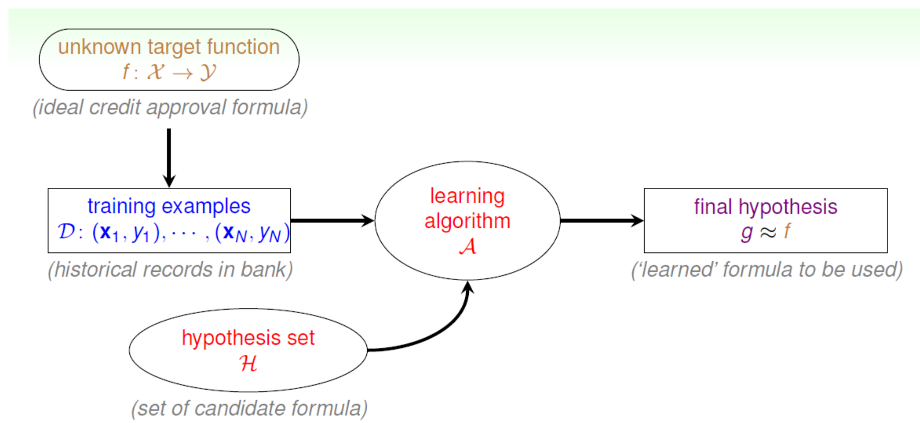

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A Simple Example (Notes)

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Outline

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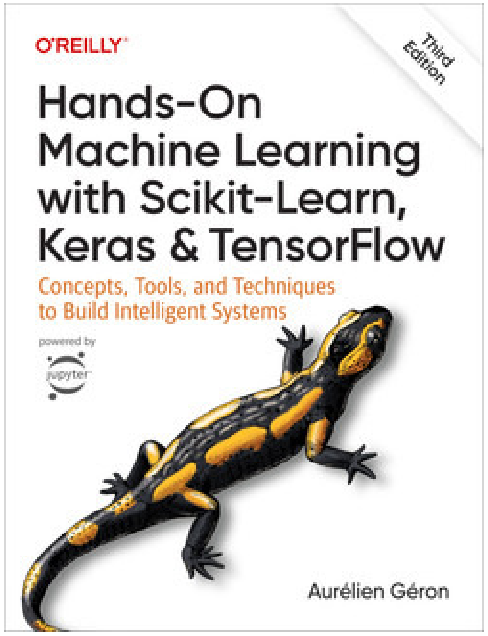

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Links (Chapter 2 in the Book)

- https://github.com/ageron/handson-ml3
- https://github.com/ageron/handson-ml3/blob/main/02_end_to_end_machine_learning_project.ipynb
- https://www.amazon.com/Hands-Machine-Learning-Scikit-Learn-TensorFlow/dp/1098125975

Main Steps

1. Look at the big picture.
2. Get the data.
3. Discover and visualize the data to gain insights.
4. Prepare the data for ML algorithms.
5. Select and train an ML model.
6. Evaluate on the test set.

Look at the Big Picture

- **Problem**: predict the median housing price in any district, given all the other metrics.
- Frame the Problem
  - What is exactly the business objective?
  - What does the current solution look like (if any)?
  - Is it supervised, unsupervised, or reinforcement learning?
  - Is it a classification task, a regression task, or something else?

Look at the Big Picture

- Select a Performance Measure
  - Root Mean Square Error (RMSE)

$$\mathrm{RMSE}(\mathbf{X}, h) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right)^2}$$

  - Mean Absolute Error (MAE)

$$\mathrm{MAE}(\mathbf{X}, h) = \frac{1}{m}\sum_{i=1}^{m}\left|h\left(\mathbf{x}^{(i)}\right) - y^{(i)}\right|$$

Look at the Big Picture

- Check the assumptions
  - It is good practice to list and verify the assumptions that have been made so far; this can help you catch serious issues early on.

Get the Data

- Download the data from the Internet
- Take a quick look at the data structure
- Create a test set

Discover and Virtualize the Data to Gain Insight

- Virtualize data
- Look for data correlations
- Experiment with attribute combinations

Prepare the Data for ML Algorithms

- Missing data (or data cleaning)
- Categorical data
- Feature scaling
- Transformation pipeline

Select and Train an ML Model

- Cross-validation
- Fine-tune the model
- Analyze the model
  - Important features

PURDUE UNIVERSITY FORT WAYNE Evaluate on the Test Set

- Do not use the test set for model selection!!!

PURDUE UNIVERSITY FORT WAYNE Summary

- Understand the relationship between AI, ML, and DL
- Become familiar with the terminology in ML
- Become familiar with the key components of ML
- Study the example of the end-to-end ML project

Q & A

Have fun!!!